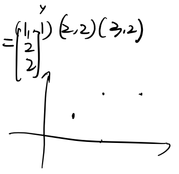
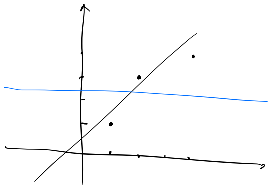
$A\vec{x} = b$ $\quad \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ $\quad (1,1)\ (2,2)\ (3,2)$

$A^TA\vec{x} = A^Tb$ $\quad m \cdot 1 + b = 1$

$\vec{x} = (A^TA)^{-1}A^Tb$ $\quad m \cdot 2 + b = 2$

$m \cdot 3 + b = 2$





$\underbrace{\qquad\qquad\qquad}_{\text{data}}$

$Xb = Y$

similarly $\quad b = (X^TX)^{-1}X^TY = \frac{1}{n\sum(x_i-\bar{x})^2} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix} \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$

$$= \begin{pmatrix} \bar{y} - \bar{x}\dfrac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} \\ \dfrac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} \end{pmatrix} = \begin{bmatrix} \hat{b}_0 \\ \hat{b}_1 \end{bmatrix}$$

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \qquad \mathbb{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \varepsilon \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{pmatrix}$$

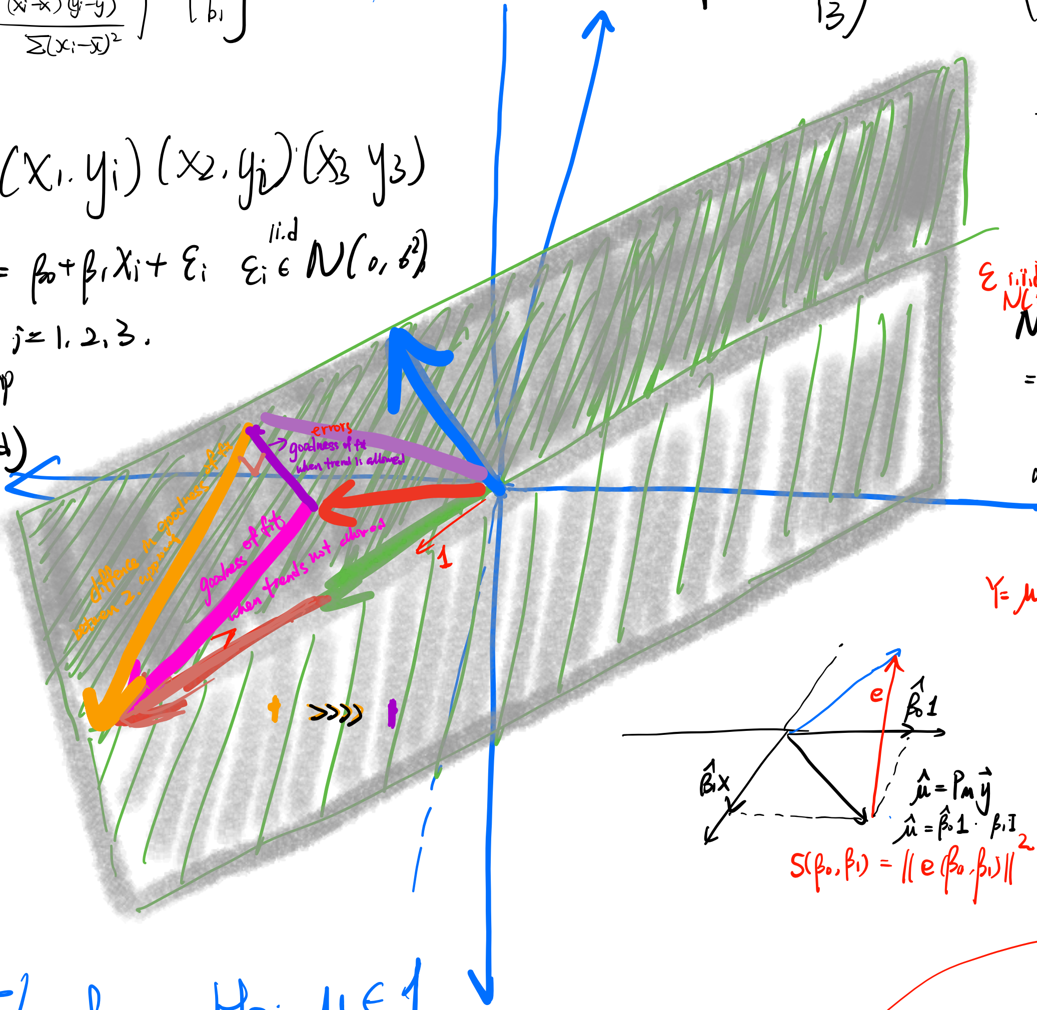
$(x_1, y_1)\ (x_2, y_2)\ (x_3\ y_3)$

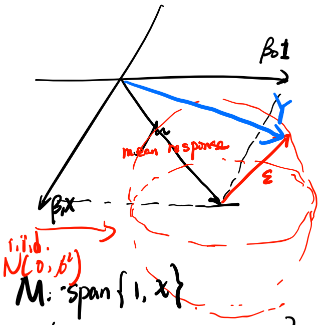
$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \varepsilon_i \overset{i.i.d}{\in} N(0, \sigma^2)$

$i = 1, 2, 3.$

i) no additional assump

ii) $\beta_1 = 0$ (no trend)


errors
goodness of fit when trend is allowed
diffence in goodness of fit between 2 approach
goodness of fit when trends not allowed
1
| >>>> |



ε i.i.d. $N(0, \sigma^2)$

M: span$\{1, x\}$

$= \{\beta_0 1 + \beta_1 x : \beta_0, \beta_1 \in R\}$

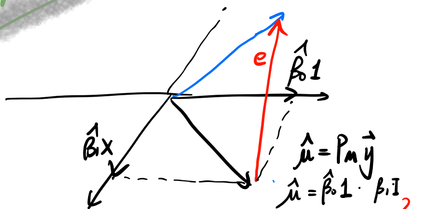
$\subseteq R^2 \subseteq R^3$

assuming x & 1 lin. ind.

$(x_1 = x_2 = x_3)$ not applied.

$Y = \mu + \varepsilon \quad \mu \in M. \quad \varepsilon \in N(0, \sigma^2 I_3)$

identity for R^3



$\hat{\mu} = P_M\vec{y}$

$\hat{\mu} = \hat{\beta}_0 1 \cdot \beta_1 I$

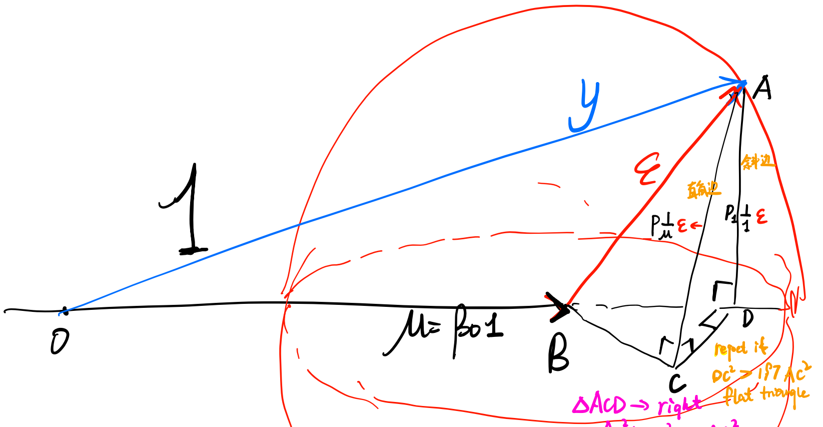
$S(\beta_0, \beta_1) = \|e(\beta_0, \beta_1)\|^2$

Therefore $\quad H_0: \mu \in \mathbb{1}$

$H_1: \mu \in M$

form D (data) D measure deviation.

from H_0 D should follow a parameter-free distribution under H_0



$\Delta ACD \to$ right

$AC^2 + CD^2 = AD^2$

residual where trend allowed · improvement · residual where no trend allowed

$\varepsilon = \vec{BA} = \vec{BD} + \vec{DC} + \vec{CA} \sim N(0, \sigma^2 I_3)$

if they're mutual orthogonal

$\vec{BD}, \vec{DC}, \vec{CA} \overset{iid}{\sim} N(0, \sigma^2)$

obvious

ANOVA Table	Sum of Squares	DoF	Distribution
Goodness of fit.	$\|P_1^\perp Y\|^2$	2	$\sigma^2 \chi^2_{(2)}$
Reduced model	$\|P_M^\perp Y\|^2$	1	$\sigma^2 \chi^2_{(1)}$
Difference	$\|P_1^\perp Y - P_M^\perp Y\|^2$	1	$\sigma^2 \chi^2_{(1)}$

Under $H_0: \mu \in \mathbb{1}$

$$F := \frac{\|P_1^\perp Y\|^2 - \|P_M^\perp Y\|^2}{\|P_M^\perp Y\|^2} = \frac{DC^2}{AC^2} \overset{d}{=} \frac{U^2}{V^2}$$

$U.V.$ i.i.d $N(0,1)$

prove. $\vec{BD} \perp \vec{DC}$

$\vec{DC} = \vec{AC} - \vec{AD}$

$\vec{BD} \cdot \vec{DC} = \vec{BD}(\vec{AC} - \vec{AD})$ ✓

$\frac{U}{V} \sim$ Cauchy

The distribution of a squared standard Cauchy is $F_{1,1}$.

Let $X \sim$ Cauchy

define $Y = X^2$.

$$P(X \le x) = \frac{1}{\pi}\arctan x + \frac{1}{2}$$

for $y \ge 0$

$$P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le x \le \sqrt{y})$$
$$= \frac{1}{\pi}\arctan\sqrt{y} + \frac{1}{2} - \frac{1}{\pi}\arctan(-\sqrt{y}) - \frac{1}{2}$$
$$= \frac{2}{\pi}\arctan(\sqrt{y}) = p$$

Set $P(Y \le y) = p$ $\quad p = \frac{2}{\pi}\arctan\sqrt{y}$

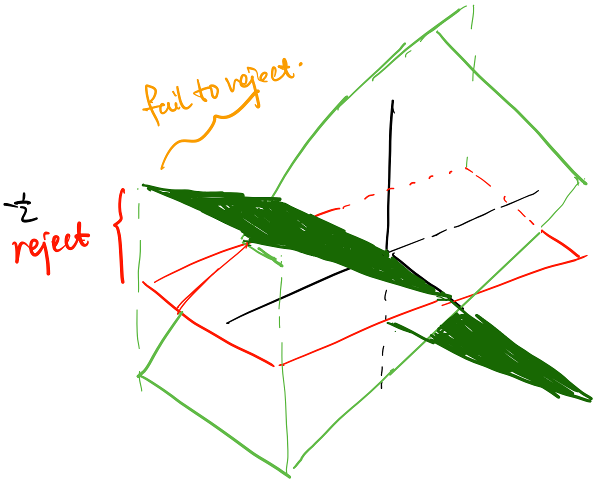
$$\rightarrow y = \tan^2(\pi p/2) \quad \text{critical value}$$

for $0 < p < 1$ $\quad$ $100p\%$ quantile $F_{1,1}$ is $\tan^2(\pi p/2)$

95%, $p = 0.95$ $\quad F_{11} = \tan^2\left(\frac{\pi \cdot 0.95}{2}\right) = 161.45$

Reject H_0: if $F := \frac{\|P_1^\perp Y\|^2 - \|P_m^\perp Y\|^2}{\|P_m^\perp Y\|^2} > F_{1,1,\alpha} = \tan^2\left(\pi(1-\alpha)/2\right)$

significance level $\alpha \in (0,1)$



Hypothesis Test: Is the observe data too far from model constrained for b

CI: which set of b give a model close enough to the data

construct CI: $H_0: \beta_1 = 0$

$\rightarrow H_0: \beta_1 = b$

$$\mathcal{L}_b = \{\beta_0 1 + bX, \ b_0 \in \mathbb{R}\}$$

$$F := \frac{\|C - D_b\|^2}{\|Y - C\|^2}$$

C: full model projection point.

D_b: projection on to the new null-plane point.

Centered predictor vector

$$X_c = X - \bar{x}1$$

$1^T X_c = 0$ $\quad 1, X_c$ orthogonal

$C = \bar{y}1 + \hat{\beta}_1 X_c$ $\quad$ mean value of y + $\hat{\beta}$ · movement of x

$D_b = \bar{y}1 + bX_c$ $\quad$ $b = 0$ in special case

$$C - D_b = (\hat{\beta}_1 - b)X_c$$

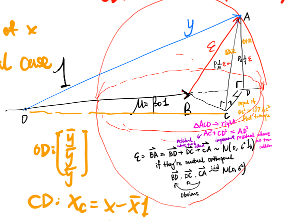
Directly CD

$$(C - D_b)^2 = (\hat{\beta}_1 - b)^2 |X_c|^2$$
$$= (\hat{\beta} - b)^2 \sum_{i=1}^{n}(x_i - \bar{x})^2 = S_{xx}$$

AC: Y−c Residual for full model $\rightarrow$

AD: Y−D residual for intercept only mode

CD: additional explained by X


y
A
ε
斜边
直角边
$P_\mu^\perp \varepsilon$
$P_1^\perp \varepsilon$
1
O
$\mu = \beta_0 1$
B
C
D
repel it
$DC^2 \to 1/7 AC^2$
flat triangle
ΔACD → right
$AC^2 + CD^2 = AD^2$
residual where trend allowed
improvement residual where no trend allowed
$\varepsilon = \vec{BA} = \vec{BD} + \vec{DC} + \vec{CA} \sim N(0, \sigma^2 I_3)$
if they're mutual orthogonal
$\vec{BD}, \vec{DC}, \vec{CA} \overset{iid}{\sim} N(0, \sigma^2)$
obvious

OD: $\begin{bmatrix} \bar{y} \\ \bar{y} \\ \bar{y} \end{bmatrix}$

CD: $X_c = x - \bar{x}1$
$= \vec{x} - \begin{bmatrix} \bar{x} \\ \bar{x} \\ \bar{x} \end{bmatrix}$

OD: $\bar{y}1$

OC: $\bar{y}1 + \hat{\beta}\left(X - \begin{bmatrix} \bar{x} \\ \bar{x} \\ \bar{x} \end{bmatrix}\right)$

$$\rightarrow F: \frac{(\hat{\beta}_1 - b)^2 S_{xx}}{\underbrace{\|Y - c\|^2/(n-2)}_{\text{residual variance}}} = \frac{(\hat{\beta}_1 - b)^2 S_{xx}^2}{s^2}$$

$F(b) \le F_{1, n-2; 0.95}$ $\quad$ 95% percentile of F-distribution

$$\rightarrow \frac{(\hat{\beta}_1 - b)^2 S_{xx}}{s^2} \le F..$$

$$|\hat{\beta}_1 - b| \le \sqrt{F_{1,n-2;0.95}} \cdot \frac{s}{\sqrt{S_{xx}}}$$

$$b \in \left[\hat{\beta}_1 \pm \sqrt{F_{1,n-2;0.95}} \frac{s}{\sqrt{S_{xx}}}\right]$$

Special. $n - 2 = 1$

$F(b) \sim F_{1,1}$

$P(F \le f) = \frac{1}{\pi}\arctan\sqrt{f} = 0.95$

$f^* = \tan^2\left(0.95\pi/2\right) = 161.45$

$\rightarrow \hat{\beta}_1 \pm \sqrt{161.45}\frac{s}{\sqrt{S_{xx}}}$

$\downarrow$

This approach directly calculate CD

Under $H_0: \mu \in 1$

$$F := \frac{\|P_1^\perp Y\|^2 - \|P_m^\perp Y\|^2}{\|P_m^\perp Y\|^2} = \frac{DC^2}{AC^2} \overset{d}{=} \frac{U^2}{V^2}$$

U, V. i.i.d $N(0,1)$

AD

$P_1 y - y \perp 1 \quad P_1 y = c1$

$(P_1 y - y)^T 1 = 0$

$(c1 - y)^T 1 = 0$

$(c1)^T 1 - y^T 1 = 0$

$$c = \frac{y^T 1}{|1 1|} = \frac{y_1 + y_2 + y_3}{(1,1,1)(1,1,1)} = \bar{y}$$

$P_1^\perp y = (y_1 - \bar{y}, y_2 - \bar{y}, y_3 - \bar{y})^T$ (no trend model error)

$$\text{Let } e_1 = \frac{1}{\|1\|} = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$e_2 = \frac{P_1^\perp x}{\|P_1^\perp x\|} = \frac{(x_1 - \bar{x}, x_2 - \bar{x}, x_3 - \bar{x})}{\| \quad \|^2}$$

$$\hat{\mu} = P_\mu y = (y^T e_1)e_1 + (y^T e_2)e_2$$

prediction

$P_M^\perp y = y - \hat{\mu}$ (trend model error)

difference = additional residual explained.

$$F: \frac{\overset{AD^2}{\|(y_1 - \bar{y}, y_2 - \bar{y}, y_3 - \bar{y})\|^2} - \overset{AC^2}{\|(y - \hat{\mu})^2\|^2}}{(y - \hat{\mu})^2}$$

OA: y

OD: $\bar{y}$

residual of null model.

residual of full

pythagorian for CD

$y \perp_m y$

residual square for the full model.

$\downarrow$

This approach use

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)}$$

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$Y = X\beta + \varepsilon$$

$$= (X^T X)^{-1} X^T (X\beta + \varepsilon)$$

$$= (X^T X)^{-1} X^T X \beta + (X^T X)^{-1} X^T \varepsilon$$

$$= \beta + (X^T X)^{-1} X^T \varepsilon \rightarrow \text{also normal}$$

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 [(X^T X)^{-1}]_{jj})$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n-p} = \frac{RSS}{n-p}$$

Therefore estimated

$$\hat{SE}(\hat{\beta}_j) = \sqrt{\sigma^2 [(X^T X)^{-1}]_{jj}}$$

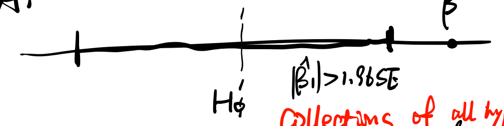
$$\approx \sqrt{\hat{\sigma}^2 [(X^T X)^{-1}]_{jj}}$$

$$T = \frac{\hat{\beta}_j - \beta_j}{\hat{SE}(\hat{\beta}_j)} \sim t_{n-p}$$

Pythagoreal for CI.

(including X)

A. Null-Centered Region

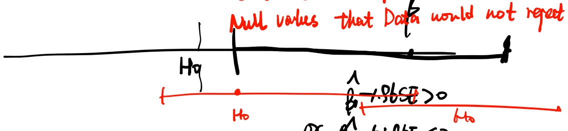


H_0

$|\hat{\beta}_1| > 1.96SE$

Collections of all hypothetical Null values that Data would not reject

B.



$\hat{\beta}_1 - 1.96SE > 0$

or $\hat{\beta}_1 + 1.96E < 0$

$\Leftrightarrow |\hat{\beta}_1| > 1.96SE$

distribution of estimator under Null

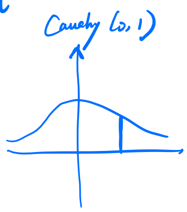
CI: plausible population parameter given the observed estimate.

Note Cauchy

$t_1 = \text{Cauchy}(0, 1)$

$$p = 2P(T_n \geq |t_{obs}|)$$

what's the prob that under the null we get as extreme as our data.



95%

$$P(-t_{0.975,\nu} \leq T \leq t_{0.975,\nu}) = 0.95$$

$$\Leftrightarrow P\left(-t_{0.975,\nu} \leq \frac{\hat{\beta}_j - \beta^{=0}}{\hat{SE}(\hat{\beta}_j)} \leq t_{0.975,\nu}\right) = 0.95$$

-1.96 $+1.96$ (non-nessary, might be bigger than 1.96.

$\Leftrightarrow \Leftrightarrow \left|\frac{\hat{\beta}}{\hat{SE}}\right|$

$$\Leftrightarrow \hat{\beta}_j - \beta \in [-1.96\hat{SE}, 1.96\hat{SE}]$$

$$\Leftrightarrow \beta - \hat{\beta}_j \in [-1.96\hat{SE}, 1.96\hat{SE}]$$

$$\Leftrightarrow \beta \in [\hat{\beta}_j - 1.96\hat{SE}, \hat{\beta}_j + 1.96\hat{SE}]$$

$$\beta = 0 \notin [\hat{\beta}_j - 1.96\hat{SE}, \hat{\beta}_j + 1.96\hat{SE}]$$

$$\Leftrightarrow |\hat{\beta}_j| > t_{0.975,n-p}\hat{SE}$$

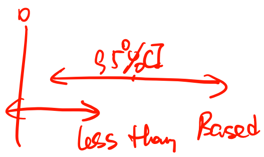
p-2side	95% CI	Interpret
$p < 0.05$	Excludes Zero	Reject $H_0 = 0$
$p > 0.05$	Contains 0.	Fail to reject $H_0 = 0$

If there's single regression coefficient: test statistic t.

If there's multiple coef. F-statistic

$H_0: \beta_1 = \beta_2 = \beta_3 = 0.$

$F_{1,n-p} = T_{n-p}^2$ special case



Less than 5% of chance that $H_0 = 0$

Based on data

⇔

⇔

	Sample Mean.	Regression Coef.
Estimator	$\bar{X}$	$\hat{\beta}_j$
true	μ	β_j
SE	$s/\sqrt{n}$	$\sqrt{\hat{\sigma}[(X^T X)^{-1}]_{jj}}$ where $\hat{\sigma} = \frac{RSS}{n-p}$
95%	$\bar{X} \pm t^* SE$	$\hat{\beta}_j \pm t^* SE$

95% CI: if we repeatedly sample data & construct interval using this procedure 95% of those intervals would contain the true coefficient.

✗ After observing data. 95% probability that the fix parameter lies in CI.

when there's only 1 predictor & intercept

$X^T X \; [X_1 - \bar{X}, X_2 - \bar{X}, \cdots X_n - \bar{X}] \begin{bmatrix} X_1 - \bar{X} \\ \vdots \end{bmatrix}$

$\hat{SE} = \frac{\hat{\sigma}}{\;}$

$$\begin{pmatrix} \vdots \\ x_n - \bar{x} \end{pmatrix} \qquad \sqrt{\sum_{i=1} (x_i - \bar{x})^2}$$